

Daily Work Report

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Work Summary

Completed Week 8: Optimization and Engineering Handover -- the final sprint of the SIA RAG Microservice project. The focus was on transforming 7 weeks of iterative development into a production-ready, fully documented, and deployable system. Implemented targeted Redis caching to address the CPU bottleneck identified in Week 7's Locust profiling. Containerized the full stack with Docker Compose including persistent LanceDB volume mounts, established a CI/CD pipeline via GitHub Actions, and produced comprehensive engineering handover documentation.

Completed Tasks

- Redis Query Caching (Targeted Optimization):

Implemented a Redis-backed cache service (cache.py) with SHA-256 key design, TTL-based expiry, and explicit cache invalidation on every document ingest. Cache hits serve in ~6ms vs ~510ms (85x improvement). Cache benchmark report produced with honest hit/miss analysis under realistic 60/40 repeated-to-novel query traffic.

- Docker Compose Stack:

Authored docker-compose.yml bringing up the FastAPI app, Redis 7, and a named persistent Docker volume for LanceDB. Configured health checks and depends_on ordering so Redis is fully healthy before the API accepts connections. Added .env.example documenting all configurable parameters including worker count and cache TTL.

- CI/CD Pipeline (GitHub Actions):

Created .github/workflows/ci.yml that runs on every PR and push: flake8 lint, black formatting check, Week 3 pytest suite, Docker image build, and a container smoke test that hits /health to verify the app starts cleanly.

- Architecture Diagrams (Mermaid):

Produced three version-controlled Mermaid diagrams: (1) System/component diagram, (2) Query request sequence diagram (cache -> embedding -> LanceDB -> extractive QA), (3) Ingestion flow diagram (PDF upload -> parse -> chunk -> embed -> store -> cache purge).

- Deployment Guide:

Authored a standalone Deployment_Guide.md assuming zero prior context, covering prerequisites, environment variables, Docker Compose steps, health verification, scaling guidance from Week 7 capacity data, and a troubleshooting section for real failure modes encountered during development.

- README Audit & Rewrite:

Completely rewrote README.md to remove all FAISS references (migrated in Week 5), reflect the current LanceDB + Redis stack, and include setup instructions, environment variables, running tests, and scaling guidance with Week 7 numbers.

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- Final Engineering Report:

Produced Final_Engineering_Report.md covering all 8 weeks with evidence-based decisions, Week 6 retrieval benchmark data, Week 7 load test numbers, and Week 8 caching results. Identifies known limitations and next steps.

- Project Presentation:

Structured a 9-slide Project_Presentation.md narrative document for non-technical stakeholders covering: problem statement, architecture, evolution (FAISS to LanceDB), retrieval bake-off, load testing findings, optimization results, and future roadmap.

Next Steps

The SIA RAG Microservice is now production-ready for a local-first deployment. The logical next step is Week 9: LLM integration -- connecting Gemini or Groq to generate synthesized natural language answers from the retrieved context, transitioning from extractive to generative RAG.